

Empirical and Molecular Formulas

Answer key with worked solutions

A. Understanding

1. Empirical vs molecular formula.

The **empirical** formula gives the simplest whole-number ratio of atoms; the **molecular** formula gives the actual number of each kind of atom in one molecule.

2. Empirical formulas.

(a) CH (b) CH₂ (c) HO (d) C₄H₉ (e) NO₂

3. What is wrong with rounding 1 : 1.50?

1.50 is not close to a whole number, so it must not be rounded. Multiply both values by 2: 2 : 3, giving A₂B₃.

B. Empirical formulas from percent composition

4. 85.63% C, 14.37% H.

C: $85.63 \div 12.01 = 7.130$ mol; H: $14.37 \div 1.008 = 14.26$ mol. Divide by 7.130: C = 1.00, H = 2.00. Empirical formula CH₂.

5. 40.04% Ca, 12.00% C, 47.96% O.

Ca: $40.04 \div 40.08 = 0.9990$; C: $12.00 \div 12.01 = 0.9992$; O: $47.96 \div 16.00 = 2.998$. Divide by 0.9990: 1 : 1 : 3. Empirical formula CaCO₃.

6. 72.36% Fe, 27.64% O.

Fe: $72.36 \div 55.85 = 1.296$; O: $27.64 \div 16.00 = 1.728$. Divide by 1.296: Fe = 1.000, O = 1.333. A ratio ending in .33 means multiply by 3: 3 : 4. Empirical formula Fe₃O₄.

C. Molecular formulas

7. CH_2O , molar mass 60.05 g/mol.

Empirical formula mass = $12.01 + 2(1.008) + 16.00 = 30.03$ g/mol. $n = 60.05 \div 30.03 = 2.00$.
Molecular formula $\text{C}_2\text{H}_4\text{O}_2$.

8. CH , molar mass 26.04 g/mol.

Empirical formula mass = 13.02 g/mol. $n = 26.04 \div 13.02 = 2.00$. Molecular formula C_2H_2 .

D. Challenge

9. 30.43% N, molar mass 92.02 g/mol.

$\text{O} = 100 - 30.43 = 69.57\%$. N: $30.43 \div 14.01 = 2.172$ mol; O: $69.57 \div 16.00 = 4.348$ mol. Ratio N : O = 1 : 2.00, so the empirical formula is NO_2 (46.01 g/mol).
 $n = 92.02 \div 46.01 = 2.00$, so the molecular formula is N_2O_4 .